Stochastic Modeling and Simulation
3rd Y - Semester 5
2025-2026

Worksheet 3

Exercice 1 *Using the Von Neumann method, starting from the seed* 0.2481*.*

1. *Generate a sequence of* 10 *random numbers uniformly distributed on* $[0; 1]$*.*

2. *Validate the generated sequence (using the three first creterions seen in the course).*

3. *Write a python programm to generate a sequence of* 100 *random numbers uniformly distributed on* $[0; 1]$*.*

   a) *Compare the empirical results with the theoretical expectations for a uniform* $U[0; 1]$ *distribution. Plot the histogram to visualize the distribution.*

   c) *Complete the validation with a* $\mathcal{X}^2$ *or Kolmogorov-Smirnov test for the uniform distribution.*

Exercice 2 *Consider the generator defined by:* $y_n = (ay_{n-1} + b) \bmod m$*, with* $a = 17, b = 43$ *and* $m = 100$*.*

1. *Generate six random numbers using the congruential method, given* $y_0 = 27$*. Comment on the resulting sequence.*

2. *Write a python programm to generate a sequence of* $n = 30$ *pseudo-random numbers normalized to* $[0; 1]$*.*

   a) *Try it out with several seeds:* $y_0 \in \{1, 27\}$*. For each seed, determine the length of the generator's period.*

   b) *Visualize the results with histograms*

Exercice 3 *Let* A *and* B *be two events, each corresponding to one of two successive independent trials:* E_A *and* E_B *respectively. Consider the events:* $A \cap B, \overline{A} \cap B, A \cap \overline{B}, \overline{A} \cap \overline{B}$*.*

Suppose $\mathbb{P}(A) = 0.3$ *and* $\mathbb{P}(B) = 0.55$*. Write the simulation algorithm to determine the realizations of these events. The objective is to find the number of occurrences during* 6 *trials (using a random number table, row by row).*

Write a python programm to generate random pairs (u_A, u_B) *and count event occurrences.*

Exercice 4 *The daily demand for a certain product has the following distribution:*

Demand	0	1	2	3	4	5
Probability	0.074	0.113	0.250	0.360	0.190	0.013

1. *Simulate the demand for the next two weeks. Random numbers should be drawn from the random number table with a precision of* 10^{-4}*.*

2. *Compute empirical mean and variance.*

Exercice 5 *Present a simulation algorithms for the following random variables:*

1. *Z Geometric distribution.*
2. *X which is the sum of 5 independent and geometrically distributed random variables with parameter $p = 0.3$. Then generate three realizations this variable.*
3. `Write a python programm to simulate` 10 `values of a random variable` X.

Exercice 6 *Present the simulation algorithms for random variables following these probability distributions:*

1. *Erlang distribution of degree k*
2. *Hyper-exponential distribution of degree 4*
3. `Write a python programm to simulate` 100 `values of an Erlang` ($k = 3, \lambda = 2$) `variable.`

Exercice 7 *Give the algorithm for simulating X a random variable whose probabilty distribution is given by*

1. *the cumulative distribution function $F(x) = \frac{1}{1+e^{-\frac{x-\alpha}{\beta}}}, \beta > 0, \alpha = \mathbb{E}[X]$.*
2. *the probability density function defined by $f(x) = \frac{2}{\pi R^2}\sqrt{R^2 - x^2}, -R \leq x \leq R$.*

`Write a python programm to simulate` 100 `values of` X `and plot the histogram.`

Exercice 8 *Write an algorithm to simulate 3 observations of a multidimensional normal distribution with mean $m = (1, 1, 2)$ and covariance matrix*

$$C = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 3 & 4 & 11 \end{pmatrix}.$$

Exercice 9 *Simulate the discrete-time Markov chain whose transition matrix is:*

$$P = \begin{pmatrix} 1/4 & 1/2 & 1/4 \\ 1/3 & 1/3 & 1/3 \\ 1/2 & 1/4 & 1/4 \end{pmatrix}$$

and initial distribution $\pi(0) = (1, 0, 0)$, using the uniform sequence : $u_0 = 0.45, u_1 = 0.87, u_2 = 0.78, u_3 = 0.65, u_4 = 0.86, u_5 = 0.18$.

Exercice 10 *A call center has two telephone lines. Calls arrive according to a Poisson process at a rate of $\lambda > 0$ calls per minute. When a call arrives, if there is an available line, it is immediately answered, otherwise it is lost. The duration of each call follows an exponential law with parameter $\mu > 0$. We want to model and simulate this system using a continuous-time Markov chain with States space : State 0: no line is busy ; State 1: one line is busy and State 2: both lines are busy.*

1. *Give the representative transition graphe and transition matrix of this Marokov chain. And present the simulation algorithm for this CTMC.*
2. `Write a python programm to simulate system evolution for` 50 `events. Estimate the proportion of lost calls (state` 2`).`